

Utathya Aich

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Researcher in Multimodal Learning · Vision-Language Models · Medical AI

RESEARCH INTERESTS

Topics: Multimodal learning, LLMs / VLMs, Agents, Medical Imaging, Representation learning.

Research Vision & Ongoing Work. My long-term goal is to build **reliable, interpretable multimodal foundation models** that combine vision, language, and structured memory. I am currently working on cross-modal alignment. Moving forward, I aim to explore:

- Unified multimodal architectures with grounded reasoning,
- Hallucination detection and information tracking in Foundation Models,
- Efficient multimodal training via low-rank, sparse, and semi-supervised approaches.

Collaborating Institutions: CMATER Lab, Jadavpur University · University of Liverpool · IIIT Bangalore · ISI Kolkata · AISC – University of South Carolina · University of Calcutta

Publications

- **Conference:** CVPRW 2026, 2 X ICPR 2026, IEEE CBMS 2026, MIUA 2025, ICDAR 2025, IJCNLP-AAACL 2025.
- **Journal:** Knowledge Based Systems, Scientific Reports, SN Computer Science

EDUCATION

- **Jadavpur University** Kolkata, India
M.Engg. in Information Technology (Specialization: Computer Vision) 2020 – 2022
 - **Master's Thesis:** *Human Pose Classification using Optimized XGBoost and SVM through TLBO optimization with Keypoints estimated from BlazePose.*
 - Advisor:** Prof. Somenath Dhibar
 - **Academic Distinction:** Graduated with **GPA: 9.11/10**; awarded the **University Gold Medal** for ranking **1st** in the department.
- **Netaji Subhash Engineering College** Kolkata, India
B.Tech in Computer Science & Engineering 2014 – 2018
 - **Bachelor's Thesis:** *Efficient Intrusion Detection in Cloud Environments using different optimization approaches.*
 - Advisor:** Prof. Partha Ghosh

PUBLICATIONS

Full Paper List

** denotes equal contribution*

Conference Papers

- *FAM-Match: Fractal-Aligned Manifold Matching with Adaptive Routing Framework for Semi-Supervised Medical Image Classification*
Gayen S., **Aich U.**, Minenkov D., Kaplun D., Sarkar R. [**CVPRW 2026**] [Link]
- *QSFL: A Quasi-Sequential Federated Learning Framework with Performance-aware Aggregation*
Aich U.*, Neogi S.*, Sengupta A.*, Bhanja H.*, Gulvanskii V., Kaplun D., Sarkar R. [**ICPR 2026**]
- *LiFGANet: Lightweight Frequency and Gradient Aware Network for Robust Image Classification*
Banerjee J.*, **Aich U.***, Bhattacharya U. [**ICPR 2026**]
- *Who Remembers What? Tracing Information Fidelity in Human-AI Chains*
Acharjee S.*, **Aich U.***, Mandal D., Ali A.* [**IJCNLP-AAACL 2025**] [Link]
- *HiLEx: Image-based Hierarchical Layout Extraction from Question Papers*
Aich U., Ghosh S., Saha T. [**ICDAR 2025**] [Link]

- *DRASU-Net: Dual-Backbone Residual Atrous Squeeze U-Net for Polyp Segmentation*
Aich U. *, Roy R. *, Eroshkin A., Kaplun D., Sarkar R. [MIUA 2025] [Link]
- *FSTS-Net: Feature-Mixing Semi-Supervised Teacher-Student Network for Polyp Segmentation*
Roy R. *, Aich U. *, Kaplun D., Sarkar R. [SIPAIM 2025] [Link]

Journal Papers

- *DAMM: Dynamic Modality Aware Weighted Embeddings Fusion for Multimodal Meme Detection*
Imam M. *, Aich U. *, Sarkar R. [Knowledge-Based Systems 2026] [Link]
- *Schizophrenia detection from electroencephalogram signals using image encoding and wrapper-based deep feature selection approach*
Aich U. *, Saha A. *, Singh P.K. [Scientific Reports 2025] [Link]
- *Efficient Brain Tumor Classification Using Filter-Based Deep Feature Selection Methodology*
Kar S. *, Aich U. *, Singh P.K. [SN Computer Science 2024] [Link]

WORK EXPERIENCE

- **CNH Industrial** Gurgaon, India
Software Developer – ML July 2022 – Present
 - **LLM & RAG Systems:** Built enterprise RAG pipelines with schema-aware retrieval and Text-to-SQL reasoning, achieving 60% latency reduction and 35% accuracy gain.
 - **Knowledge Graph Reasoning:** Developed Neo4j-based knowledge-graph reasoning pipelines, reducing manual support workload by 50%.
 - **Geospatial Computer Vision:** Implemented GEO-SAM-based segmentation for automated field-boundary detection, enabling large-scale geospatial analytics.
 - **Applied Research:** Mentored 5+ Interns and 3+ Students Under CSR activity
- **(VARSHA) Asymptotic Technologies Pvt. Ltd.** Kolkata, India
Co-founder – Entrepreneurship Nov 2018 – Aug 2020
 - **Real-Time Medical Dispatch System:** Built geospatial allocation and routing engine for **VARSHA**, an Uber-style ambulance/on-call doctor service, reducing response times by **28%**.
 - **Pilot Deployment:** Ran pilot across local regions, coordinating logistics with clinical partners and supporting **50+** emergency cases.

ACADEMIC SERVICE

- **Organizing Challenge:**
 - * **ICPR 2026** – VISTAC 2
 - * **ICME 2026** – ECHO
 - * Co-organizer of **ComSys Hackathon 6 (2025)** – “Hack the Fake, Save the Real: Deepfake Audio Classification in the Wild.”
 - * Co-organizer of **ComSys Hackathon 5 (2025)** – “Face Identification under Different Environmental Conditions.”
- **Reviewer:**
 - * **Conference:** CBMS 2026, ICIP 2026, IEEECAI 2026, ICWSM 2026, 7th CICBA 2025, ComSys 2025, ComSys 2024.
 - * **Journal:** Information Fusion, PRAI

AWARDS & HONOURS

- **Academic Achievements:**
 - * **University Gold Medalist (2022)** – Highest Academic Performance (M.Tech) in the Information Technology Department, Jadavpur University.
 - * Ranked **1st out of 20,000+** candidates in **PGET 2020 (MAKAUT Post Graduate Exam)**.
 - * Qualified **GATE CSE 2020** with national scholarship for Master’s study.
- **Professional Recognition:**
 - * **Employee of the Quarter – Q3 (2024)** – CNH Industrial.
 - * **Spot Award – Employee of the Month (2024)** – Data Analytics & AI Team, CNH Industrial.
 - * Winner, **Innovation Sprint 2023** – semi-automated CV annotation tool, CNH Industrial.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **Deep Learning:** PyTorch, TensorFlow, HuggingFace Transformers
- **LLM / RAG Stack:** LangChain, LlamaIndex, VectorDBs, Neo4j, FastAPI
- **MLOps & Infrastructure:** Databricks, MLflow, Spark, Azure, CI/CD

RESEARCH COLLABORATORS & MENTORS

- **Prof. Ram Sarkar** — Professor, Jadavpur University
- **Prof. Tulika Saha** — Assistant Professor, IIIT Bangalore & University of Liverpool
- **Prof. Ujjwal Bhattacharya** — Professor, Indian Statistical Institute, Kolkata
- **Dr. Swarnendu Ghosh** — Marie Curie Postdoctoral Fellow, University of Ljubljana, Slovenia
- **Dr. Pawan Kumar Singh** — Assistant Professor, Jadavpur University